

Mihir Upadhyay

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EDUCATION

New York University

M.S. in Data Science

- **Course Assistant**, Responsible Data Science
- **Assistant**, Courant Institute of Mathematical Sciences

New York, US

Sep 2024 - Present

Jan 2025 - Present

Nov 2024 - Present

Indian Institute of Technology BHU

B.Tech in Chemical Engineering

Varanasi, India

Jul 2017 - May 2021

WORK EXPERIENCE

Flixstock

Technical Product Manager (Generative AI)

Gurugram, India

Oct 2022 - Sep 2023

- Directed FlixLite's image catalog revamp with StyleGAN2 & DensePose-RCNN, enhancing image quality and catalog size by 20%.
- Optimized ML pipelines with distributed computing, reducing errors by 30% and runtime by 60% to improve scalability.

Standard Chartered Bank

AI/ML Analyst

Bengaluru, India

Jul 2021 - Sep 2022

- Developed ARIMA-based monitoring system, boosting credit card fraud detection efficiency by reducing false positives by 30%.
- Built a portal with RESTful APIs using Flask to decommission redundant tools, saving \$60K annually in costs for 45 VMs.

Data Science Intern

May 2020 - Jul 2020

- Trained an XGBoost model with 96% accuracy on transaction data, boosting fraud detection and reducing false positives by 25%.
- Developed Tableau dashboards for KPIs like transaction volume and response time, reducing reporting time by 4 hours/week.

PUBLICATIONS

"When Should We Orchestrate Multiple Agents?"

Submitted to ICML 2025 (under review)

Umang Bhatt, Sanyam Kapoor, Katherine M. Collins, Ilia Sucholutsky, **Mihir Upadhyay**, Francesco Quinzan, Adrian Weller, Andrew Gordon Wilson, Muhammad Bilal Zafar

PROJECTS

Professor Ratings Analysis with Statistical Insights, Hypothesis Testing, and Predictive Modeling @ NYU

Nov 2024

- Analyzed professor ratings using Mann-Whitney U, Kolmogorov-Smirnov, and Levene's tests, identifying significant gender-based differences with a p -value < 0.005 .
- Developed a polynomial regression model to predict professor ratings from tags (R-squared: 0.85) and a logistic regression model for attractiveness ratings, achieving 85% accuracy and an AUC-ROC of 0.92 using feature engineering and SMOTE for class imbalance.
- Performed Kruskal-Wallis tests to analyze the impact of subjects on ratings, identifying the top 3 subjects with the highest mean ranks and visualizing results using bar plots and heatmaps.

Enhanced Human-Background Image Synthesis using BLIP2 and DensePose @ Flixstock

Aug 2023

- Developed an ML pipeline using DensePose for anatomical extraction and BLIP2 for scene understanding, achieving 95% accuracy in anatomical placement and scene consistency.
- Improved realism with shadow correction (40% lighting consistency) and deployed the pipeline as a Flask API with React front-end, reducing response time to under 20 seconds.
- Optimized AWS S3 image storage and enhanced the pipeline to support 100+ concurrent requests with sub-200ms latency.

LLM-Optimized Relationship Manager Portal with Adaptive Query Resolution @ Standard Chartered Bank

Sep 2022

- Developed an LLM-powered chatbot (GPT-3) with fine-tuned retrieval-augmented generation (RAG) and intent classification, reducing query resolution time by 40%.
- Built a continuous learning pipeline using user feedback loops and embeddings-based similarity search, enhancing chatbot adaptability to evolving financial queries with a 35% improvement in contextual relevance.
- Integrated WebSocket-based messaging for seamless RM-client communication, reducing manual intervention by 50%.
- Implemented a secure RM portal using Spring, Java, React.js, PostgreSQL, and JWT authentication, enabling portfolio tracking, personalized service promotion, and RM performance monitoring, increasing engagement by 30%.

SKILLS

Programming: Python, SQL, PROC SQL, PySpark, C/C++, Shell/Bash, HTML, JavaScript

Frameworks: Pandas, Numpy, NLTK, PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, LangChain, Hugging Face, Spark

Databases: MS SQL Server, MySQL, PostgreSQL, Oracle RDBMS, DynamoDB

Tools: AWS (S3, SageMaker Airflow), GCP, Kubernetes, Looker, Tableau, PowerBI, Git, AEM, Prolific, Pavlovia